

Project Plan Proposal

Eye Snake Localization Using Vision

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1. Introduction

Vitreoretinal surgery demands sub-millimeter precision, posing significant challenges for surgical tool localization inside the eye. The I²RIS (Integrated Intraocular Robotic System) snake robot, developed at JHU LCSR, is a promising platform for performing delicate procedures on the retina. However, accurate real-time localization of the snake robot's gripper tip remains an open problem critical for enabling autonomous or semi-autonomous surgical assistance.

This project proposes a vision-based localization system for the eye snake robot that leverages deep learning to predict the 3D position of the robot gripper from camera imagery and point cloud data captured during surgery. The system is trained in simulation using the AMBF (Asynchronous Multi-Body Framework) environment and is intended to transfer to real hardware through domain adaptation and fine-tuning.

2. Goals and Significance

2.1 Clinical Need

Vitreoretinal procedures require sub-millimeter accuracy to avoid damage to the delicate retinal tissue. Current approaches often rely on kinematic models from joint encoders alone, which can accumulate errors due to cable slack, hysteresis, and calibration drift in the snake robot's cable-driven mechanism. A vision-based localization approach can provide an independent, high-accuracy position estimate that either supplements or replaces encoder-based tracking.

2.2 Project Goals

The primary goal is to develop and validate a vision-based localization pipeline for the I²RIS snake robot that achieves clinically useful accuracy (target: <5 mm, stretch goal: sub-millimeter) in both simulation and on real hardware. Specific objectives include:

- Build a comprehensive simulation dataset using AMBF with diverse snake robot configurations and ground-truth gripper positions
- Implement and compare multiple deep learning architectures across RGB, point cloud, and multimodal fusion approaches
- Collect and annotate real robot data via stereo camera and ROS integration
- Perform sim-to-real transfer with domain adaptation and fine-tuning
- Deploy a real-time inference system integrated into the ROS surgical control pipeline

3. Technical Approach

3.1 Simulation Environment

The project leverages the AMBF simulation framework, an open-source platform developed at JHU LCSR for real-time surgical robotics simulation. A pre-existing I²RIS snake robot simulation environment provides the foundation for synthetic data generation. The simulator renders the eye interior with the snake robot in various configurations, producing synchronized RGB images, depth maps, and point clouds along with ground-truth gripper and base positions in 3D.

Each training sample consists of a 1024×1024 RGB image, a corresponding point cloud of 4096 3D points, and ground-truth labels including gripper position (mm), base position (mm), and rotation (radians). The dataset comprises 150 samples with the snake robot visible in diverse poses within the simulated eye.

3.2 Model Architectures

Six model architectures have been implemented and evaluated, spanning three input modalities:

3.2.1 Point Cloud Models

PointNet: Processes raw 3D point sets using shared MLPs and a global max-pooling operation to produce a fixed-size feature vector for regression.

PointNet++: Extends PointNet with hierarchical feature learning through set abstraction layers that capture local geometric structure at multiple scales.

DGCNN: Applies dynamic graph convolution (EdgeConv) to learn local and global point cloud features by constructing k-nearest-neighbor graphs in feature space.

3.2.2 RGB Model

HRNet-Heatmap: Uses High-Resolution Network with a soft-argmax heatmap decoder to predict 2D keypoint locations from RGB images, which are then lifted to 3D using camera intrinsics and depth information.

3.2.3 Multimodal Fusion Models

Hybrid Fusion: Combines ResNet-18 image features, PointNet point cloud features, and joint encoder readings through an MLP-based fusion head for 3D regression.

Transformer Fusion: Uses a Vision Transformer (ViT) for image encoding and PointNet for point cloud encoding, fusing both via cross-attention transformer layers before final regression.

3.3 Preliminary Results

All six architectures have been trained and evaluated on 15 held-out test samples from the 150-sample AMBF simulation dataset. The results are summarized below:

Approach	Modality	Method	Error (mm)	Std (mm)	<5mm	<10mm
HRNet-Heatmap	RGB	HRNet+Softmax	2.98	1.47	93.3%	100%
Hybrid Fusion	RGB+PC+Joint s	ResNet-18+PointNet+MLP	3.49	1.23	93.3%	100%
Transformer Fusion	RGB+PC	ViT+PointNet+Cross-Attention	3.88	1.32	80.0%	100%
DGCNN	Point Cloud	CNN+EdgeConv	4.84	1.52	53.3%	100%
PointNet++	Point Cloud	Hierarchical PointNet	8.97	2.76	6.7%	60.0%
PointNet	Point Cloud	Shared MLP+Max pooling	9.49	1.88	0.0%	60.0%

Table 1. Comparison of localization approaches on simulation test set (15 samples).

The HRNet-Heatmap approach achieves the best mean error at 2.98 mm, with 93.3% of predictions within 5 mm. The Hybrid Fusion model shows comparable accuracy with a lower standard deviation (1.23 mm), suggesting more consistent predictions. Pure point cloud methods (PointNet, PointNet++) underperform significantly, likely due to the small dataset size and the challenge of localizing a small tool within a large scene point cloud. DGCNN bridges the gap with its graph-based feature learning.

3.4 Remaining Technical Work

Significant work remains to transition from simulation results to a deployable real-robot system:

- **Real data collection:** Setting up ROS integration for synchronized stereo camera capture from the physical I²RIS system, including stereo calibration for depth reconstruction.
- **Sim-to-real transfer:** Fine-tuning the best-performing simulation model on real data, with intrinsics conditioning to handle the difference between fixed simulation camera parameters and variable real camera setups.
- **Shadow-based depth:** Investigating shadow cues as an additional depth signal for monocular setups where stereo is unavailable.
- **Real-time optimization:** Model pruning, quantization, or architecture simplification to achieve inference speeds suitable for surgical control loops.
- **Failure case handling:** Detecting and recovering from localization failures (e.g., occlusion, out-of-distribution poses) with confidence estimation.

4. Deliverables

4.1 Minimum Deliverables (Must Have)

1. Working simulation model with validated localization accuracy
2. ROS integration for synchronized real-robot data capture
3. Stereo camera calibration for the physical setup
4. Annotated real data collection from the I²RIS robot
5. Fine-tuned model achieving measurable accuracy on real robot data
6. Real-time inference capability
7. Camera intrinsics generalization across different camera configurations
8. Documented sim-to-real performance gap analysis

4.2 Stretch Goals (Nice to Have)

- Sub-millimeter accuracy on real robot
- Sensor fusion combining vision-based localization with joint encoder readings
- ICRA 2027 paper submission

5. Project Timeline

The project spans 15 weeks with bi-weekly mentor meetings (Monday and Wednesday) and continuous progress updates via Microsoft Teams. Phases overlap where feasible to maintain momentum.

Phase	Weeks	Activities
Phase 1: Foundation	W1–W3	AMBF simulation setup, data collection pipeline, baseline model training
Phase 2: Model Development	W4–W7	Implement and evaluate 7 model architectures, comparative analysis
Phase 3: Real Data Collection	W6–W9	ROS setup, stereo calibration, real robot data collection and annotation
Phase 4: Sim-to-Real Transfer	W9–W12	Domain adaptation, model fine-tuning, intrinsics conditioning, validation
Phase 5: Deployment	W12–W15	ROS real-time integration, live demo, documentation and final report

Table 2. Project timeline with phase descriptions and week ranges.

Note that Phases 2–3 and Phases 3–4 overlap intentionally: real data collection can begin while model evaluation is being finalized, and transfer learning experiments can start as soon as initial real data is available.

6. Project Management

Meetings: Bi-weekly meetings with mentors on Mondays and Wednesdays, with micro-updates communicated to Prof. Adnan Munawar via Microsoft Teams.

Collaboration: The project welcomes collaborators (contact: czhuang3@jh.edu). Collaboration opportunities exist in real data collection, model implementation, and ROS integration.

Tools and Infrastructure: PyTorch for model development, OpenCV for image processing, ROS for robot communication, NumPy and Matplotlib for data analysis and visualization. Training is performed on GPU-equipped workstations with the AMBF simulator running on Ubuntu.

Risk Mitigation: The primary risk is sim-to-real domain gap exceeding expectations. This is mitigated by the multi-model comparison approach (allowing selection of the most transfer-friendly architecture), domain randomization in simulation, and incremental fine-tuning strategies. Hardware access delays are mitigated by front-loading simulation work in Phases 1–2.

7. References

- [1] He et al., "Deep Residual Learning for Image Recognition," CVPR 2016.
- [2] Qi et al., "PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation," CVPR 2017.
- [3] Sun et al., "Deep High-Resolution Representation Learning for Visual Recognition," CVPR 2019.
- [4] Munawar et al., "A Real-Time Dynamic Simulator and an Associated Front-End Representation Language," IROS 2019.
- [5] I²RIS Robot: Integrated Intraocular Robotic System, JHU LCSR.
- [6] Wang et al., "Dynamic Graph CNN for Learning on Point Clouds," ACM TOG, 2019.
- [7] Vaswani et al., "Attention Is All You Need," NeurIPS 2017.
- [8] Luvizon et al., "Human Pose Regression by Combining Indirect Part Detection and Contextual Information," BMVC 2017.
- [9] Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," ICLR 2021.
- [10] Qi et al., "PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space," NeurIPS 2017.